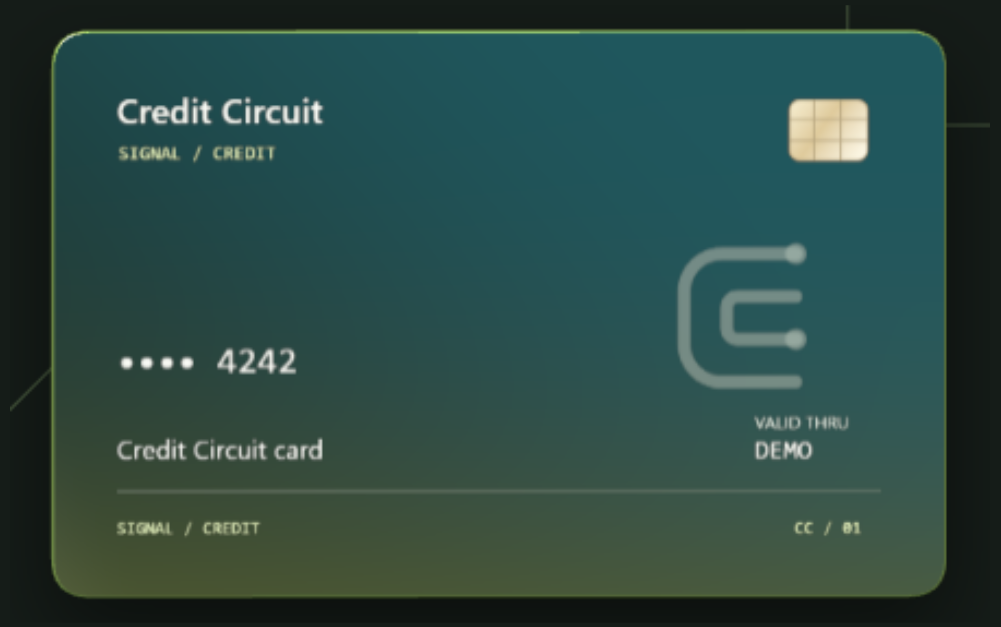
Marvens Jean / UCI 2123

Credit Circuit.

Every purchase starts a signal.

Credit card transaction simulator



My background is in retail.

Development interested me because I wanted to understand how everyday processes work.

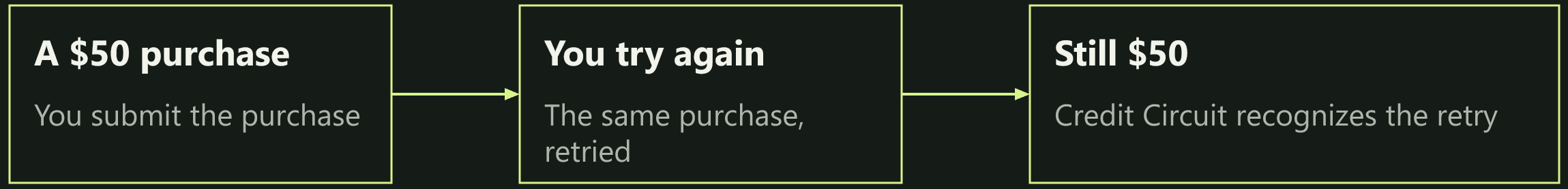
My interest in finance led me to explore credit card transactions.



PROBLEM AND SOLUTION

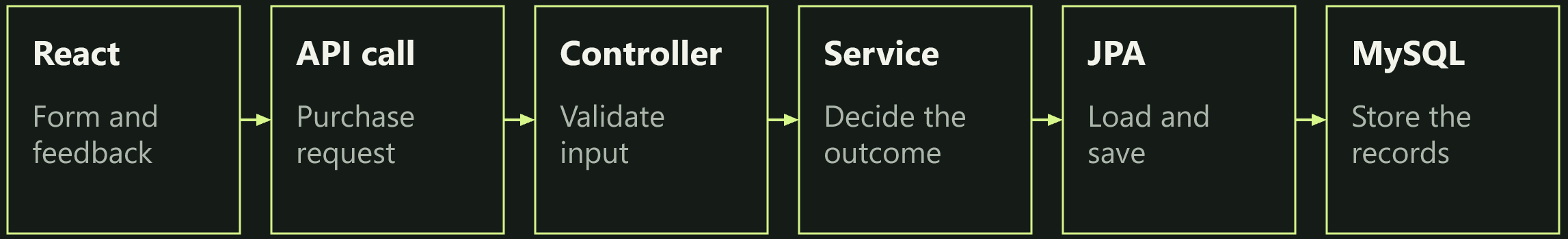
Trying again shouldn’t cost you twice

A dropped connection shouldn’t turn a $50 purchase into a $100 problem.



Credit Circuit See what happened to every purchase, with clear decisions and refunds you can trace.

The stack



One backend. Four tables.

Each part has a clear job, so I can follow a purchase through the app.

The purchase service

01 Check the customer and lock their account

02 Check the assigned card

03 Look for the same request ID

04 Check expiry, account status, and credit

05 Approve or decline

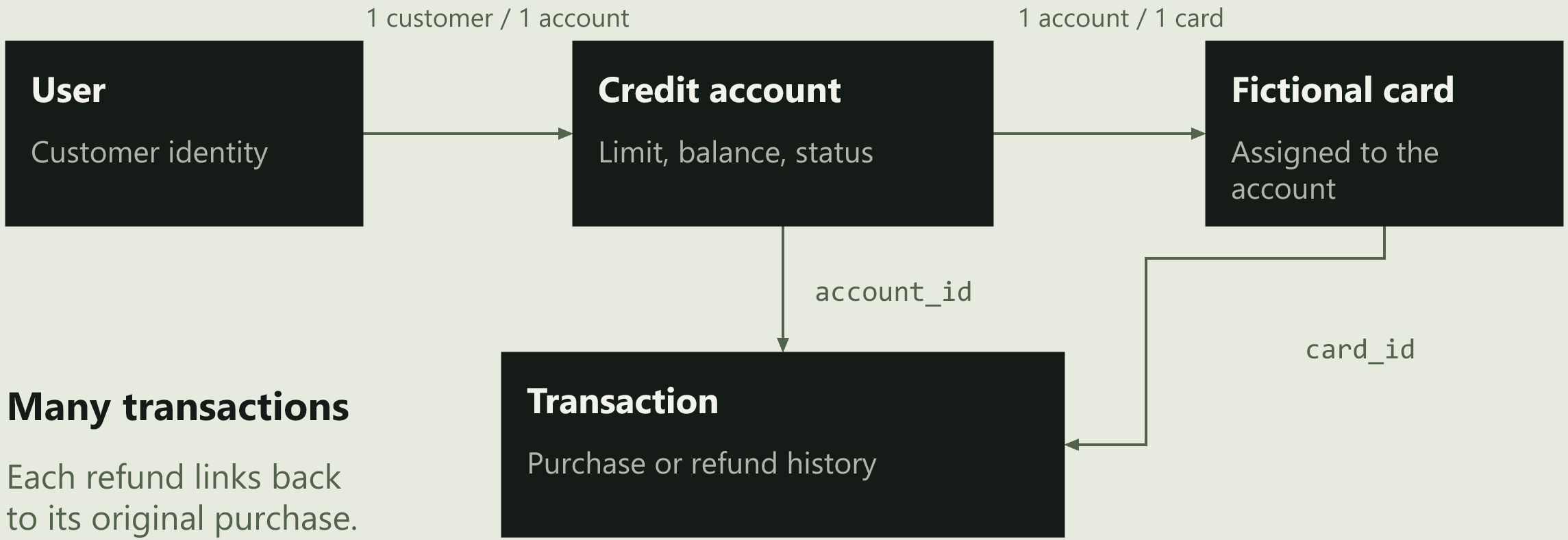
06 Save the outcome and balance together

TransactionService

The service connects each history record to the checked account and assigned card.

```
transaction.setAccount(
    customerAccount);
transaction.setCard(
    assignedCard);
```

The relationships



Many transactions

Each refund links back to its original purchase.

The request and result

One submitted form

Amount and merchant
Assigned card details
Request ID

One returned response

Transaction outcome
Current account balance

---

201 New outcome, including a decline

200 Identical retry

FAILURE PATHS

When something goes wrong

Declined purchase	Save the reason. Leave the balance unchanged.
Same request again	Return the existing transaction.
Same ID, changed details	Reject the conflicting request.
Concurrent spending or failed write	Lock the account. Commit or roll back together.

Security and testing

95.56%

Java line coverage

47 Java tests

Including 13 real MySQL integration tests

Recorded runs: October 9, 2026

BCrypt passwords

Signed JWTs in React memory
Role and ownership checks

Postman

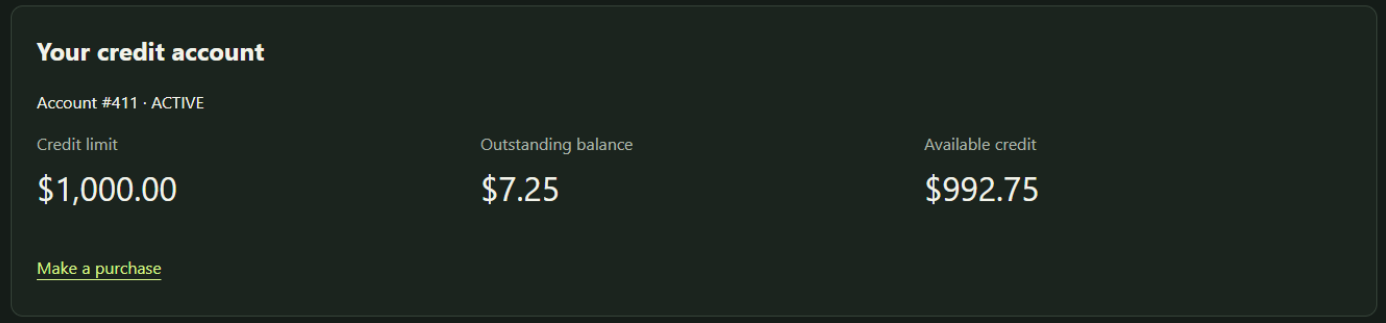
56 requests / 83 assertions passed

SonarQube quality gate passed

RECORDED DEMO

A purchase in Credit Circuit

Credit at a glance



Fictional cards and balances

01 Dashboard and flippable card

02 $50 approved purchase

03 Decline above available credit

04 History and full refund

05 Admin account freeze

LESSONS LEARNED

What this project taught me

Consistency starts with the design

The balance and history commit together.
Every refund points to its original purchase.

Failure paths deserve their own tests

Retries, concurrent spending, and failed writes
need evidence from the database tests.

Thank you